

**POST GRADUATE DIPLOMA IN
COMPUTER APPLICATIONS
[PGDCA-NEW]**

Term-End Examination

June, 2025

**MCS-206 : OBJECT ORIENTED
PROGRAMMING USING JAVA**

Time : 3 Hours

Maximum Marks : 100

Weightage : 70%

Note : *Question No. 1 is compulsory and carries
40 marks. Attempt any **three** questions
from the rest.*

1. (a) Discuss the utility of 'ByteStream' class
in Java with suitable example. 5

- (b) Explain the use of getter methods and setter methods in Java. Give suitable example code for each. 5
- (c) How is event handling performed in JavaFx ? Explain with the help of a program. 5
- (d) Describe the term 'package' in Java. Explain the accessibility rules for packages in Java. 5
- (e) What are access specifiers ? List and discuss the various access specifiers available in Java. 5
- (f) Write a java program which takes a string as input and displays its length. Support your code with suitable

comments. Also print the first character of the string. 5

(g) What is a constructor ? How are parameterised constructors defined in Java ? Explain with suitable code. 5

(h) Discuss the features of 'AWT' and 'Swing' used for GUI programming in Java. 5

2. (a) What is JDBC ? Describe the utility of JDBC drivers. Write the steps to connect a database through JDBC. Also, write a suitable code in Java to showcase the database connectivity using JDBC. 10

- (b) What is 'Inheritance' ? Give advantages of inheritance. Also, write a suitable code in Java to describe the implementation of inheritance in Java. Support your code with suitable comments. 10
3. (a) What is an exception in Java ? Give basic causes of exception. Write a program to explain how to handle 'ArrayIndexOutOfBoundsException' in Java. Also, explain the use of 'finally' clause in exception handling. 10
- (b) What is multithreading in Java ? Explain how thread synchronization is achieved in Java, with the help of a

program. Support your program with suitable comments. 10

4. (a) Compare 2-tier architecture with 3-tier architecture in relation to JDBC. Draw suitable block diagram for both the architectures. Also, discuss the functionality of core components of JDBC. 10
- (b) Explain, why main method in Java is always static. 5
- (c) Write a Java program using appropriate GUI components, which takes your name and e-mail id as input and displays them on click of a button. Support your code with suitable comments. 5

5. Write short notes on the following : 5×4=20

- (a) Decision constructs in Java
- (b) PreparedStatement Interface
- (c) Role of Hashset and Hashmap
- (d) Assertion statement and its significance

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No. of Printed Pages : 6

MCS-206

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December, 2025

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Time : 3 Hours

Maximum Marks : 100

Weightage : 70%

Note : *Question No. 1 is compulsory and carries
40 marks. Attempt any **three** questions
from the rest.*

1. (a) What is encapsulation ? How is encapsulation implemented in Java ?
Explain through a program. 5
- (b) Differentiate between abstract class and interface in Java. Give example of each. 5
- (c) Discuss the utility of map interface in Java, with the help of a program. 5
- (d) What is object serialization in Java ?
Explain the working and utility of object serialization. 5
- (e) Describe the utility of 'final' and 'super' keywords in Java with suitable code. 5

- (f) Write a Java program to find the sum of two matrices of dimension 3×3 . Support your code with suitable comments. 5
- (g) Explain the use of ResultSet interface in JDBC, with the help of a program. 5
- (h) Explain the advantage of function overloading in Java with the help of a suitable program. 5
2. (a) What is multithreading ? Write suitable code in Java to illustrate the creation of thread. Also, explain the use of `isAlive()` method in multithreaded programming with the help of a program. Support your code with suitable comments. 10

- (b) What is Number Class ? Explain the use of parseInt (String s) method with the help of an example. 5
- (c) Explain event handling in Java FX, with the help of an example. 5
3. (a) Describe the term 'Exception Handling' in Java. Discuss the role of 'throws' clause in exception handling, with the help of a suitable code in Java. Support your code with suitable comments. Also, explain the advantages of exception handling. 10
- (b) Write a program in Java using JDBC to store the employees

information which includes ID, Name, Date of Birth, Department, position in a database table named Employee. Support your code with suitable comments. 10

4. (a) What is Polymorphism ? Explain different types of polymorphism that are supported by Java, with the help of example codes for each. 10
- (b) Discuss the purpose of stream classes in Java. How do byte streams differ from character streams ? Give suitable example for each. 10
5. Write short notes on the following : $4 \times 5 = 20$
- (a) CallableStatement Interface

- (b) Garbage collection and role of finalize()
method in garbage collection
- (c) JavaFX and its features
- (d) Utility of TreeSet and HashSet

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No. of Printed Pages : 3

MCS-206

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Time : 3 Hours

Maximum Marks : 100

Note : Question No. 1 is compulsory. Attempt any *three* questions from the rest.

1. (a) What is a class ? How a class is defined in Java ? Explain the differences between public and private classes. 5
- (b) What is a relational operator ? List and explain the use of different relational operators in Java. 5
- (c) Write a Java program which take radius of a circle as input and display its area. Define appropriate class and methods in the program. 5

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MCS-206

- (d) What is inheritance ? How it is implemented in Java ? Define Vehicle class and inherit classes “Two wheeler vehicle” and “four wheeler vehicle” from vehicle class. Define appropriate constructors in all these three classes. 10
 - (e) What is File class in Java ? Explain the use of any two methods in File class. 5
 - (f) What is assertions in Java ? Explain how assertions are enabled and disabled in Java. 5
 - (g) Explain the use of JDBC in two-tier system. Also, explain the need of JDBC driver. 5
2. (a) Write a Java program to explain the use of “switch case” and “break” statements. 5
 - (b) What is method overloading ? Implement it with the help of a program ? 5
 - (c) What is abstract class ? List its advantages. In java how an abstract class is defined ? Can you crate object from an abstract class ? Also, explain differences between abstract class and interface. 10
3. (a) What is an exception ? Discuss the Throwable class hierarchy. Also, explain checked and unchecked exception.

- (b) What is an access specifier (control) ? What are the different types of access specifiers ? Explain why methods of interface are public ? Write Java code to show how interface is extended. 10
4. (a) What are the various steps for connecting Java application with database using JDBC ? Explain with the help of program segment. Also, explain the difference between commit and rollback.
- (b) What is the need of event handling in GUI programming ? Explain how events are handled in Java Fx. 10
5. Write short notes on the following : 4×5=20
- (a) Multithreading in Java
 - (b) Use of “finally” block in Exception Handling
 - (c) uses of “super” keyword
 - (d) Arrays in Java

No. of Printed Pages : 6

MCS-206

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Time : 3 Hours

Maximum Marks : 100

Weightage : 70%

Note : (i) *Question No. 1 is compulsory.*

(ii) *Attempt any **three** questions from the
rest.*

1. (a) What is object oriented programming ?

Explain the concept of encapsulation with the help of an example. 5

- (b) What is a variable ? Explain the use of a “final” variable with the help of a Java program. 5

- (c) What is JDBC ? Briefly explain the core components of JDBC. Write and explain necessary steps to create a JDBC application. 10

- (d) What is an object in Java ? Explain how an object is assigned to another object, with the help of Java code. 5

(e) What is Java thread model ? Explain thread life cycle in Java with the help of a diagram. 7

(f) What is a package ? Explain how you can create your own package in Java. What are the different types of packages in Java ? 8

2. (a) What is constructor ? Explain the need of parameterised constructor in Java program with the help of an example. 5

(b) What is an Exception ? What are the different types of Exceptions ? Explain the need of exception handling in Java. Also,

explain how exceptions, which are “checked by compiler”, are handled with the help of an example. 10

- (c) Write a Java program which takes a string as input and displays its reverse. Also, find its length. 5

3. (a) What is collection-API in Java ?

What is Date Class ? Explain the use of *compareTo* and *after* methods of Date class. 5

- (b) What is HashSet class ? Explain its *two* constructors. 5

- (c) What is inheritance ? List and explain the types of inheritances supported by Java.

Write a Java program to illustrate implementation of inheritance. 10

4. (a) Explain the differences between “ByteStream” and “CharacterStream.” 5

- (b) What is AWT ? Explain the differences between AWT and Swing. 5

- (c) What is “ResultSet” in JDBC programming ?
What are the different types of ResultSet ?
Explain ResultSet Holdability with the help of an example. 10

5. Write short notes on the following : $4 \times 5 = 20$

(a) JavaFX

(b) Polymorphism

(c) Java Interface

(d) User-Defined Exceptions

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No. of Printed Pages : 4

MCS-206

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PROGRAMMING USING JAVA**

Time : 3 Hours

Maximum Marks : 100

Weightage : 70%

Note : (i) *Question No. 1 is compulsory and carries 40 marks.*

(ii) *Attempt any **three** questions from the rest.*

1. (a) Briefly discuss the concept of static binding with suitable example. 5
- (b) Illustrate the execution of switch-case statement and if-then-else statement, with suitable code in Java for each. 5

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- (c) What are constructors ? How do constructors differ from methods ? 5
 - (d) Compare Multiple Inheritance and Multi-level Inheritance. Give suitable Java code and block diagram for each. 5
 - (e) What is an Interface in Java ? How Interface differs from abstract class ? 5
 - (f) What is the significance of assertion statement in Java ? How usage of assertion statement differs from the procedure of exception handling in Java ? 5
 - (g) Differentiate between static and default methods in interfaces in Java. 5
 - (h) What is multithreading concept in Java ? How does it differ from the concept of multiprocessing ? 5
2. (a) Differentiate between the following : 10
- (i) Map Interface and HashMap Class
 - (ii) TreeSet and HashSet
 - (iii) Set and Map
 - (iv) HashSet and HashMap
 - (v) List and Set

- (b) Discuss the purpose of stream classes in Java. How do byte streams differ from character streams ? Give suitable example for each. 10
3. (a) What is purpose of AWT-API in Java ? Briefly discuss the role of component class, container class, panel class, windows class and frame class of AWT. 10
- (b) Compare the 2-tier architecture of JDBC with 3-tier architecture of JDBC, draw suitable block diagram for both architectures. Also, discuss the functionality of core components of JDBC. 10
4. (a) What is the purpose of Connection Interface in JDBC ? Briefly discuss and compare statement interface, Prepared-StatementInterface and CallableStatement interface. 10
- (b) Briefly discuss the concept of Garbage collection in Java. Give advantages and disadvantages of Garbage collection. Also discuss the role of finalize() method in Garbage collection. 10

5. (a) Write a program in Java to print multiplication table of the number entered by the user. Support your code with suitable comments. 5
- (b) Compare the concept of Throw and Throws in Java with the help of example code. 5
- (c) What is JDBC Batch Processing ? Explain with the help of a program. 10

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Time : 3 Hours

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Note : (i) *Question No. 1 is compulsory and carries 40 marks.*

(ii) *Attempt any **three** questions from the rest.*

1. (a) Briefly discuss the concept of dynamic binding with a suitable example. 5
- (b) Illustrate the execution of while loop and do---while loop with suitable code in Java for each. 5

- (c) What are access specifiers ? Briefly discuss the purpose of public, private, protected and default access specifiers. 5
- (d) Compare Hierarchical inheritance with Multiple inheritance. Give suitable pseudo code and block diagram for each. 5
- (e) Differentiate between overloading and overriding. Give suitable example for each. 5
- (f) What are annotations in Java ? Briefly discuss the purpose of the following, Built-in Java annotations : 5
- (i) @ Override
 - (ii) @ Suppress Warning
 - (iii) @ Deprecated
- (g) Differentiate between class and interface in Java. 5
- (h) Briefly discuss the various states of thread life cycle. Also draw block diagram for thread life cycle. 5
2. (a) Write short notes on the following :
- $2\frac{1}{2} \times 4 = 10$
- (i) Closable Interface
 - (ii) Flushable Interface
 - (iii) Autoclosable Interface
 - (iv) I/O Exception

- (b) What is a vector class ? How does vector class differ from stack class ? Briefly discuss any *two* methods of both. 10
3. (a) What are swings in Java ? How does Java swings differ from Java AWT ? Also, list the features of JavaFx that makes it superior to Java AWT and JavaSwings. 10
- (b) What is the purpose of JDBC driver ? Briefly discuss the Type-I, Type-II, Type-III and Type-IV of JDBC drivers. 10
4. (a) What is a transaction in DBMS ? How is it maintained in the JDBC applications ? How do dirty read, non-repeatable reads and phantom read makes the database inconsistent ? What mechanism can we use to resolve these inconsistencies ? 10
- (b) Differentiate between the following : 5+5
- (i) JDBC RowSet and ResultSet
- (ii) JavaDataObject API and JDBC API
5. (a) Write a program in Java to print the name of week days, (MON to SUN) when the user enters the number between 1 to 7. Support your code with suitable comments.

- (b) Compare the concepts of Final, Finally and Finalize in Java with example. 6
- (c) Explain the use of “Super” keyword in Java with the help of an example. 5
- (d) Write Java program to demonstrate how a thread is created in Java. Also set priority of created thread to B. Support your program with suitable comments. 5

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Time : 3 hours

Maximum Marks : 100

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Note : Question no. 1 is **compulsory** and carries 40 marks. Answer any **three** questions from the rest.

1. (a) What are access specifiers ? Explain the difference between public and private access specifiers in Java, with the help of an example. 6
- (b) Write a Java program which takes a string as input and displays its length. 5
- (c) Write a Java program to find the sum of two matrices of dimension 3×3 . 5

- (d) What is a package in Java ? Explain accessibility rules for packages. 7
- (e) Explain the use of Map class in Java, with the help of a program. 5
- (f) What is object serialization ? Explain the working and use of object serialization. 6
- (g) Explain event handling in JavaFX, with the help of a program. 6
- 2.** (a) What is inheritance ? Explain the advantages of inheritance. Explain order of calling of constructors in the case of multilevel inheritance, with the help of a program. 10
- (b) Write a program using JDBC to store student information which includes name, roll number, date of birth and programme code. Give suitable comments in your program. 10
- 3.** (a) What is multithreading ? Explain how thread synchronization is achieved in Java, with the help of a program. 10
- (b) What is polymorphism ? Explain the different types of polymorphism that are supported by Java, with the help of examples. 10

4. (a) Explain how Java provides platform independence, with the help of a diagram. 5
- (b) Explain the use of “final” and “super” keywords of Java. 5
- (c) What is exception ? Explain basic causes of exception. Write a program to explain how to handle “ArrayIndexOutOfBoundsException” in Java. Also explain the use of “finally” clause in exception handling. 10
5. (a) What is encapsulation ? How is it implemented in Java ? Explain through a program. 5
- (b) Explain the use of getter methods and setter methods in Java programming, with the help of an example. 5
- (c) Explain the difference between abstract class and interface. 5
- (d) Explain the use of “ByteStream” class in Java. 5
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Note : (i) *Question No. 1 is compulsory and carries
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rest.*

1. (a) Write a java program which takes three numbers as input and displays average of these numbers. 5

- (b) Briefly explain use of the following logical operators with the help of example of each : 6
- (i) <
 - (ii) >
 - (iii) > =
- (c) Explain function overloading with the help of a program. 6
- (d) Explain use of abstract class with the help of a program. Explain why abstract class cannot be instantiated. 6
- (e) What is constructor ? Explain how parameterised constructors are defined in Java. 6
- (f) Explain use of ResultSet interface in JDBC programming with the help of a program. 5
- (g) Write a Java program to display table of a given number between 1 to 20. 6
2. (a) What is inheritance ? Explain advantages of inheritance. Explain how inheritance in Java is implemented with the help of a program. Also explain the advantages of method overriding. 10

- (b) What is multithreading ? Briefly explain how thread is created in Java. Explain use of `isAlive( )` method in multithreaded programming with the help of a program.

10

3. (a) Write a Java program which takes a string as input and displays reverse of the input string.

5

- (b) Explain different uses of “final” keyword of Java.

5

- (c) Explain advantages of exception handling. Explain how exceptions are handled in Java, with the help of a program. Also explain use of “throws” clause in Java.

10

4. (a) Explain use of interface in Java with the help of an example program. Also describe difference between abstract class and interface.

10

- (b) Discuss the utility of set and vector classes in Java Programming with the help of a program.

10

5. (a) What is GUI programming in Java ?
Briefly explain the features of “AWT” and
“Swing” in Java. 5
- (b) Write a Java program using appropriate
GUI components, which takes your name
and e-mail id as input and displays them
on click of a button. 7
- (c) What is JDBC ? What is the use of JDBC
driver ? Explain the steps to connect a
database through JDBC. 8